

Table S1: Model GWL Crossing Years and Storyline Assignments

Model	Scenario	GWL +2C Year	GWL +3C Year	Winter SL (+2C)	Summer SL (+2C)	Winter SL (+3C)	Summer SL (+3C)
ACCESS-CM2	SSP5-8.5	2039	2056.0	Increasing Frequency	Increasing Frequency	Decreasing Frequency	Neutral / Other
ACCESS-ESM1-5	SSP5-8.5	2040	2061.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
BCC-CSM2-MR	SSP5-8.5	2044	2066.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
CESM2	SSP5-8.5	2037	2056.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
CESM2-WACCM	SSP5-8.5	2034	2054.0	Neutral / Other	Neutral / Other	Increasing Frequency	Increasing Frequency
CMCC-CM2-SR5	SSP5-8.5	2034	2053.0	Increasing Frequency	Increasing Frequency	Decreasing Frequency	Increasing Frequency
CMCC-ESM2	SSP5-8.5	2040	2056.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
CNRM-CM6-1	SSP5-8.5	2041	2059.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
CNRM-CM6-1-HR	SSP5-8.5	2030	2052.0	Increasing Frequency	Increasing Frequency	Neutral / Other	Neutral / Other
CNRM-ESM2-1	SSP5-8.5	2046	2065.0	Decreasing Frequency	Neutral / Other	Decreasing Frequency	Decreasing Frequency
CanESM5	SSP5-8.5	2023	2041.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
E3SM-1-0	SSP5-8.5	2033	2048.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
EC-Earth3	SSP5-8.5	2036	2058.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
EC-Earth3-CC	SSP5-8.5	2037	2057.0	Decreasing Frequency	Neutral / Other	Neutral / Other	Decreasing Frequency
EC-Earth3-Veg-LR	SSP5-8.5	2042	2062.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
GFDL-ESM4	SSP5-8.5	2053	2076.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
HadGEM3-GC31-LL	SSP5-8.5	2031	2048.0	Increasing Frequency	Decreasing Frequency	Increasing Frequency	Decreasing Frequency
HadGEM3-GC31-MM	SSP5-8.5	2035	2050.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Neutral / Other
IITM-ESM	SSP5-8.5	2049	2077.0	Neutral / Other	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
INM-CM4-8	SSP5-8.5	2047	2070.0	Decreasing Frequency	Decreasing Frequency	Increasing Frequency	Increasing Frequency
INM-CM5-0	SSP5-8.5	2046	2075.0	Increasing Frequency	Increasing Frequency	Neutral / Other	Increasing Frequency
IPSL-CM6A-LR	SSP5-8.5	2035	2051.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
KACE-1-0-G	SSP5-8.5	2024	2044.0	Neutral / Other	Increasing Frequency	Increasing Frequency	Increasing Frequency
KIOST-ESM	SSP5-8.5	2039	2065.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
MIROC-ES2L	SSP5-8.5	2048	2071.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
MIROC6	SSP5-8.5	2054	2077.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
MPI-ESM1-2-HR	SSP5-8.5	2050	2074.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
MPI-ESM1-2-LR	SSP5-8.5	2049	2072.0	Decreasing Frequency	Decreasing Frequency	Increasing Frequency	Increasing Frequency
MRI-ESM2-0	SSP5-8.5	2039	2065.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
NorESM2-MM	SSP5-8.5	2055	2077.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
UKESM1-0-LL	SSP5-8.5	2032	2047.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
ACCESS-CM2	SSP2-4.5	2041	2072.0	Decreasing Frequency	Increasing Frequency	Increasing Frequency	Decreasing Frequency

Model	Scenario	GWL +2C Year	GWL +3C Year	Winter SL (+2C)	Summer SL (+2C)	Winter SL (+3C)	Summer SL (+3C)
ACCESS-ESM1-5	SSP2-4.5	2046	nan	Increasing Frequency	Increasing Frequency	Neutral / Other	Neutral / Other
CESM2	SSP2-4.5	2044	2081.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
CESM2-WACCM	SSP2-4.5	2040	2076.0	Increasing Frequency	Decreasing Frequency	Increasing Frequency	Increasing Frequency
CMCC-CM2-SR5	SSP2-4.5	2039	2067.0	Neutral / Other	Increasing Frequency	Decreasing Frequency	Decreasing Frequency
CMCC-ESM2	SSP2-4.5	2041	2066.0	Increasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
CanESM5	SSP2-4.5	2025	2050.0	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
EC-Earth3	SSP2-4.5	2045	2086.0	Neutral / Other	Decreasing Frequency	Decreasing Frequency	Decreasing Frequency
EC-Earth3-CC	SSP2-4.5	2036	2074.0	Decreasing Frequency	Neutral / Other	Decreasing Frequency	Decreasing Frequency
EC-Earth3-Veg-LR	SSP2-4.5	2050	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
GFDL-ESM4	SSP2-4.5	2074	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
HadGEM3-GC31-LL	SSP2-4.5	2034	2062.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
IITM-ESM	SSP2-4.5	2062	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
INM-CM4-8	SSP2-4.5	2064	nan	Increasing Frequency	Increasing Frequency	Neutral / Other	Neutral / Other
INM-CM5-0	SSP2-4.5	2073	nan	Neutral / Other	Neutral / Other	Neutral / Other	Neutral / Other
IPSL-CM6A-LR	SSP2-4.5	2034	2065.0	Neutral / Other	Neutral / Other	Decreasing Frequency	Increasing Frequency
KACE-1-0-G	SSP2-4.5	2024	2051.0	Increasing Frequency	Increasing Frequency	Increasing Frequency	Increasing Frequency
KIOST-ESM	SSP2-4.5	2041	nan	Neutral / Other	Increasing Frequency	Neutral / Other	Neutral / Other
MIROC-ES2L	SSP2-4.5	2064	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
MIROC6	SSP2-4.5	2074	nan	Increasing Frequency	Neutral / Other	Neutral / Other	Neutral / Other
MPI-ESM1-2-HR	SSP2-4.5	2064	nan	Increasing Frequency	Increasing Frequency	Neutral / Other	Neutral / Other
MPI-ESM1-2-LR	SSP2-4.5	2058	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
MRI-ESM2-0	SSP2-4.5	2050	nan	Decreasing Frequency	Decreasing Frequency	Neutral / Other	Neutral / Other
NorESM2-MM	SSP2-4.5	2079	nan	Increasing Frequency	Increasing Frequency	Neutral / Other	Neutral / Other
UKESM1-0-LL	SSP2-4.5	2034	2060.0	Decreasing Frequency	Neutral / Other	Increasing Frequency	Increasing Frequency